# WorldForge 2 - Open Issues and Release-First Improvement Plan

Written: 2026-10-15
Status: ACTIVE. This is the current top priority for all work in this repo.
Source: a /court trial of four claims (12 independent juror agents per claim, prosecution, defense, judge), each finding then checked against the repository.
Verdicts: docs/court/2026-10-15/
PDF copy: docs/IMPROVEMENT_PLAN.pdf (regenerate with: python scripts/plan_to_pdf.py)

## 1. The verdicts

Score = jurors who voted FOR the claim, out of 12, scaled to 10.

- Claim 1 - "Worth continuing in its current direction": GUILTY 11-1. Score 1/10.
- Claim 2 - "Ready (or close) to sell": GUILTY 12-0. Score 0/10.
- Claim 3 - "Generator + deep campaign in one app is the right scope": GUILTY 9-3. Score 2.5/10.
- Claim 4 - "Tauri + Rust tiles + SQLite is the right foundation": NOT GUILTY 9-3. Score 7.5/10.

Bottom line, the same in all four rulings: the engine is sound, the ORDER OF WORK is wrong. Nobody argued for a rewrite or a new stack. The project keeps adding simulation layers while nothing has shipped, no outsider has used it, and the parts a buyer would touch first (install, saved files, memory on their machine) are unproven.

Redesign question: the court found no case for redesigning the program. The only "new idea" it supports is a reframing of the PRODUCT, not the code: sell it as a science-grounded worldbuilding map tool, with the living campaign as a second mode, and let testers decide how deep the campaign goes.

## 2. Open issues

***I-1 No evidence anyone wants it [severity: critical]***

- No users, testers, buyers, price, sales page or waitlist anywhere.
- All 12 jurors on claim 2 rested their vote on this.

***I-2 Nothing can be installed or bought [critical]***

- No LICENSE file, no release tags, version still 0.1.0 in package.json and Cargo.toml.
- README only documents dev mode (npm run tauri dev, run.bat).
- CI (.github/workflows/ci.yml) type-checks, tests and runs vite build, but never produces a Tauri installer.

***I-3 Features are being added before anything ships [critical]***

- Sept-Oct 2026 went into Living World rows 02-10 (face genomes, councils, leisure and games, masterworks) and a UI redesign.
- Each new layer adds code for one person to maintain and new economy side effects (see I-9).

***I-4 Saved worlds and campaigns [high]***

- FIXED for new saves: slice 1 of docs/TWO_APPS_AND_FILE_UPLOAD_PLAN.md (section 8) stopped save_world_as and new_campaign from deleting the world's human layer. Gate: campaign_key_split_tests.
- OWNER DECISION 2026-10-15: old saves are NOT supported. Every file carries SAVE_FORMAT (SQLite user_version); older files are refused with a clear message.
- Slice 4 (sibling-file pairing) BUILT 2026-10-15. STILL OPEN: slices 3 and 5 (.wf2 bundle, drag-and-drop).
- FIXED 2026-10-15: round-trip test of current saves (commands::save_roundtrip_tests). Found and fixed: every save with an unreachable route pair could not be reopened (base_days INFINITY -> JSON null).

***I-5 Peak memory has never been measured [high]***

- SCOREBOARD.md: "Time is benchmarked; memory is not, and memory is the likelier failure on a customer's machine."
- The year-30 out-of-memory crash (unbounded history journal) is recorded as fixed, but no long full-size campaign has been measured since.
- All 9 jurors who acquitted the architecture said they would flip if memory turns out bad.

***I-6 Campaign speed at real scale [medium]***

- CLAUDE.md section 5.5: at ~1,200 hubs the trade phase is still ~8,300-10,200 ms per simulated year after the house_for cache (down from 18-26 s).
- A decade-long campaign on a big world is minutes of waiting. Fine for testers, worth fixing before a paid release.

***I-7 Product identity is undecided [high]***

- Still open: one app, or a world generator plus a campaign player?
- There is no one-sentence pitch. The Designer juror: "a stranger could not say in five seconds whether this is a map generator or a merchant-and-dynasty sim".

***I-8 Legal basics for a paid product [high before any sale]***

- No commercial licence or end-user agreement (EULA).
- No audit of dependency licences (Tauri, React, PixiJS, rusqlite and the rest of the npm/cargo tree).
- No check of data rights: the Koppen-Geiger reference map used by the climate harness, the commodity-history text in the Goods Codex, any bundled fonts or images.
- Most commits (728 of 988) were written by an AI agent. Check how your jurisdiction treats copyright in AI-assisted code before selling.
- This list is not legal advice. Get a short professional review before the first sale.

***I-9 Quality debt [medium]***

- The frontend (~33k lines per SCOREBOARD, ~60k TS/TSX in total) has zero tests.
- Rust <-> TypeScript types are mirrored by hand (types/campaign.ts), so a renamed field fails silently.
- The top-10% wealth share has fallen out of its band, cause "not yet explained" (SCOREBOARD 2026-09-30b). The bisect is queued.
- Tests have been red on main at times (inheritance gate, unrest_topples_councils). Check their current state in CI.
- There are 116 git branches. Prune the merged ones.

***I-10 Local test environment [low]***

- FIXED 2026-10-15: build.rs now embeds the Common-Controls manifest for every target; local `cargo test` launches again.

## 3. The plan

Work the phases in order. Each item has a "done when" so progress is checkable. Tick items here ([x]) as they land, and regenerate the PDF.

***Phase 0 - Feature freeze (now)***

- **DONE** Freeze written into CLAUDE.md section 0: no new Living World / people-layer / campaign features until v0.1 has shipped to testers. Bug fixes, performance, save safety and release work are allowed.
- **DONE** Living World rows: owner decided 2026-10-15 — LEAVE AS-IS, frozen on main, not extended.

***Phase 1 - Save safety (I-4)***

- **DONE** REPLACED by owner decision: no old-save support. Save-format stamp (SAVE_FORMAT in user_version) + refusal of older files; fixture removed.
- **DONE** Regression test: commands::save_roundtrip_tests (new world + campaign saved, reopened in a fresh app, resumes at tick 30, runs a tick, reloads its streamed blob; an unstamped file is refused without touching the open world).
- **DONE** Slice 4: open_world returns sibling_campaign; App offers to load it. Mismatch warning now names the world.
- **TODO** Optional, after testers: slice 3 (.wf2 bundle) and slice 5 (drag-and-drop).

***Phase 2 - Measure (I-5, I-6)***

- **DONE** Target: 8 GB RAM Windows laptop; budget 3 GB peak working set (MEM_BUDGET_MB in real_world_diagnostics.rs; change there if the owner picks another).
- [~] Profiled 3600x1800 to year 12 (owner stopped it: enough evidence). Harness: real_world_diagnostics::measure_full_size_campaign on a cached mock world. Peak ~2.2-2.4 GB after fixes; 40-60 s per simulated year. Owner target: 200 years — speed is now the blocker, not memory.
- **DONE** SCOREBOARD 2026-10-15 + 2026-10-16.
- **DONE** Growth found: transient whole-sim copies in advance_resilient (3 -> 2 copies) + autosave as one giant String (now streamed). Steady memory is not climbing. Claim 4 stands.
- [~] Speed for 200 years: recent_trades ring buffer shipped (-15%); FxHash tried and reverted (negative result). 2026-10-16b: carriage data restructured, bit-identical (arrival calendar tick/transit.rs, single-entry trade log, per-day route cache) -> 17-20 s per simulated year, flat through year 12, i.e. ~65-70 min for 200 years (SCOREBOARD 2026-10-16b). OPEN (owner): is ~1 hour for 200 years acceptable? Next costs if not: dispatch ranking scan (~4 s/yr), pre-trade consumption/prices (~3 s/yr), relay hops (~6 M/yr, a carriage-design question).

***Phase 3 - Decide the product (I-7)***

- **DONE** One app, map first (owner, 2026-10-15). Recorded at the top of docs/TWO_APPS_AND_FILE_UPLOAD_PLAN.md.
- **DONE** Pitch written as the headline of README.md.
- **DONE** First buyers: fantasy authors, TTRPG GMs, worldbuilding hobbyists (map + export); sim players for the campaign mode.

***Phase 4 - Legal basics (I-8)***

- **DONE** Closed commercial licence chosen; LICENSE added (EULA.md still to draft).
- **DONE** docs/THIRD_PARTY_LICENSES.md via `node scripts/license_report.mjs`: 0 npm flags, 5 MPL-2.0 crates via Tauri (OK unmodified).
- **DONE** docs/DATA_RIGHTS.md: Koppen + elevation fixtures are test-only (never shipped); commodity text is original; fish plates are AI-generated (check tool terms); no bundled fonts.
- [~] EULA.md drafted (marked DRAFT). STILL OPEN: one professional review of it + the AI-code question (docs/DATA_RIGHTS.md lists the open points).

***Phase 5 - Ship v0.1 to testers (I-1, I-2)***

- **DONE** .github/workflows/release.yml: tauri build on windows-latest, NSIS installer uploaded as artifact (run by hand or on a v* tag). Built locally: WorldForge 2_0.1.0-beta.1_x64-setup.exe, 9 MB.
- [~] Version is 0.1.0-beta.1 (MSI dropped: WiX rejects "-beta"). STILL OPEN (owner): push tag v0.1.0-beta.1 → draft release with the installer; check it, then publish.
- **DONE** README "Install (testers)" section: download, SmartScreen note, backups.
- **TODO** Recruit 5-10 outside worldbuilders. Ask each to: install, make a world, export a map, save, reopen, start a campaign, run 10 years.

- **TODO** Collect what broke, what they used, what they ignored, and whether they would pay (and how much).
- **TODO** Put up a one-page waitlist or pre-order page with a stated price, and count sign-ups.

***Phase 6 - Let the testers steer***

- **TODO** Only now decide whether the campaign / people layers grow, stay, or get cut, based on what testers actually used.
- **TODO** Re-run /court on claims 1-3 with the tester evidence in the case file.

***Housekeeping (any time, small)***

- **TODO** Fix or explain the red tests on main, bisect the top-10% wealth share drift.
- **TODO** Prune merged branches (116 now).
- **DONE** Rust -> TS check: scripts/check_types.mjs (no new dependency), run in CI after tsc. Matches 275 TS types to same-named Rust serde structs and fails if TS declares a field Rust never sends (a renamed/removed field). 0 drift today after allowlisting 7 client-side fields. Limits: field NAMES only (not types); a TS type with no same-named Rust struct is not checked; panel smoke tests still not written.

## 4. What NOT to do

- Do not rewrite the stack. Claim 4 was acquitted: every recorded failure was in application code, not in Tauri, Rust, tiles or SQLite.
- Do not split into two codebases before testers have used it.
- Do not start new Living World rows, new goods, new institutions or new panels until Phase 5 is done.
- Do not treat "the tests pass" as "it is ready". The tests measure the code, not demand.

## 5. How to use and update this file

- Every new Claude Code session: read CLAUDE.md section 0 first, then this file. Tell the owner which phase is current and which items are open before starting other work.
- If the owner asks for a new feature while Phase 0-5 items are open, remind them of the freeze and of the next open item, then do what they decide.
- When an item lands: tick it here, add evidence (commit, test name, number), and regenerate the PDF: `python scripts/plan_to_pdf.py`
- When all of Phase 5 is ticked, the freeze is over. Update CLAUDE.md section 0.